Rosser's theorem: consistent axiomatizable extensions of Q are incomplete

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Editorial note. The fixed-point equation labelled RT inside the proof span is part of the counted span; the fixed-point lemma behind it is declared as a standard prerequisite.

Editorial note. The comparative remark citing the first incompleteness theorem is navigational only.

Editorial scope. One original theorem of the unit's Rosser section, counted once: incompleteness for consistent axiomatizable extensions of Q via the Rosser trick. The counted theorem and its author proof are retained in full; the section's motivational discussion before the proof is not counted.

Allowed standard prerequisites (editorial).

- A consistent axiomatizable theory T extending Q, and the Rosser derivability predicate, which says that there is a proof of the sentence in T and no shorter refutation of it, built from the proof predicate Prf_T and the refutation predicate Refut_T, both represented in Q.
- The bounded-quantifier collapse available in Q (the less-nsucc lemma of the representability unit): from Q-proofs of not-phi(k) for each k below n, Q proves the bounded universal statement for all z below the numeral of n.
- The fixed-point lemma for Q, giving a sentence R_T with Q proving R_T iff not Rosser-provable of its own Godel number.
- The fact that in a consistent T no k below the Godel number of an actual derivation can be the Godel number of a derivation of the negated sentence.

Source reference. Original lemma fixed-point: sources/olp-incompleteness-incompleteness-provability/pinned-original-source.tex, source lines 616–616. The proof invokes the fixed-point lemma to obtain the Rosser sentence. It is a navigation pointer to material the counted proof cites; the counted proof retains its own argument in full. The lemma is proved in the same unit and is published as corpus card X2-incompleteness-incompleteness-provability-fixed-point.

Source reference. Original theorem first-incompleteness: sources/olp-incompleteness-incompleteness-provability/pinned-original-source.tex, source lines 518–518. The

proof contains one comparative remark pointing at the omega-consistent version of the first incompleteness theorem. It is a navigation pointer to material the counted proof cites; the counted proof retains its own argument in full.

Theorem 0.1. *Let* ThT *be any consistent, axiomatizable theory extending* ThQ. *Then* ThT *is not complete.*

Proof. Recall that Prov[ThT](y) is defined as $\exists x$ Prf[ThT](x, y), where Prf[ThT](x, y) represents the decidable relation which holds iff x is the Gödel number of derivation of the sentence with Gödel number y. The relation that holds between x and y if x is the Gödel number of a *refutation* of the sentence with Gödel number y is also decidable. Let not(x) be the primitive recursive function which does the following: if x is the code of a formula A, not(x) is a code of $\neg A$. Then Refut[ThT](x, y) holds iff Prf[ThT](x, not(y)). Let Refut[ThT](x, y) represent it. Then, if Th$T \vdash \neg A$ and δ is a corresponding derivation, Th$Q \vdash$ Refut[ThT](gn δ, gn A). We define Prov[ThT](y) as

$$\exists x\,(\mathrm{Prf}[\mathrm{Th}T](x, y) \wedge \forall z\,(z < x \to \neg\,\mathrm{Refut}[\mathrm{Th}T](z, y))).$$

Roughly, Prov[ThT](y) says "there is a proof of y in ThT, and there is no shorter refutation of y." Assuming ThT is consistent, Prov[ThT](y) is true of the same numbers as Prov[ThT](y); but from the point of view of *provability* in ThT (and we now know that there is a difference between truth and provability!) the two have different properties. If ThT is *in*consistent, then the two do *not* hold of the same numbers! (Prov[ThT](y) is often read as "y is Rosser provable." Since, as just discussed, Rosser provability is not some special kind of provability—in inconsistent theories, there are sentences that are provable but not Rosser provable—this may be confusing. To avoid the confusion, you could instead read it as "y is shmovable.")

By the fixed-point lemma, there is a formula $R_{\mathrm{Th}}T$ such that

$$\mathrm{Th}Q \vdash R_{\mathrm{Th}}T \leftrightarrow \neg\,\mathrm{Prov}[\mathrm{Th}T](\mathrm{gn}\,R_{\mathrm{Th}}T). \tag{1}$$

In contrast to the proof of first-incompleteness , here we claim that if ThT is consistent, ThT doesn't derive $R_{\mathrm{Th}}T$, and ThT also doesn't derive $\neg R_{\mathrm{Th}}T$. (In other words, we don't need the assumption of ω-consistency.)

First, let's show that Th$T \vdash\!/\, R_{\mathrm{Th}T}$. Suppose it did, so there is derivation of $R_{\mathrm{Th}T}$ from T; let n be its Gödel number. Then Th$Q \vdash$ Prf[ThT]($\overline{n}$, gn $R_{\mathrm{Th}T}$), since Prf[ThT] represents Prf[ThT] in ThQ. Also, for each $k < n$, k is not the Gödel number of derivation of $\neg R_{\mathrm{Th}T}$, since ThT is consistent. So for each $k < n$, Th$Q \vdash \neg\,$Refut[ThT]($\overline{k}$, gn $R_{\mathrm{Th}T}$). By ?? , Th$Q \vdash \forall z\,(z < \overline{n} \to \neg\,\mathrm{Refut}[\mathrm{Th}T](z, \mathrm{gn}\,R_{\mathrm{Th}T}))$. Thus,

$$\mathrm{Th}Q \vdash \exists x\,(\mathrm{Prf}[\mathrm{Th}T](x, \mathrm{gn}\,R_{\mathrm{Th}T}) \wedge \forall z\,(z < x \to \neg\,\mathrm{Refut}[\mathrm{Th}T](z, \mathrm{gn}\,R_{\mathrm{Th}T}))),$$

but that's just Prov[ThT](gn $R_{\mathrm{Th}T}$). By 1 , Th$Q \vdash \neg R_{\mathrm{Th}T}$. Since Th$T$ extends ThQ, also Th$T \vdash \neg R_{\mathrm{Th}T}$. We've assumed that Th$T \vdash R_{\mathrm{Th}T}$, so Th$T$ would be inconsistent, contrary to the assumption of the theorem.

Now, let's show that Th$T \vdash\!/\, \neg R_{\mathrm{Th}T}$. Again, suppose it did, and suppose n is the Gödel number of derivation of $\neg R_{\mathrm{Th}T}$. Then Refut[ThT](n, gn $R_{\mathrm{Th}T}$) holds, and since Refut[ThT] represents Refut[ThT] in ThQ, Th$Q \vdash$ Refut[ThT]($\overline{n}$, gn $R_{\mathrm{Th}T}$).

We'll again show that ThT would then be inconsistent because it would also derive $R_{\mathrm{Th}T}$. Since

$$\mathrm{Th}Q \vdash R_{\mathrm{Th}T} \leftrightarrow \neg\,\mathrm{Prov}[\mathrm{Th}T](\mathrm{gn}\,R_{\mathrm{Th}T}),$$

and since ThT extends ThQ, it suffices to show that

$$\mathrm{Th}Q \vdash \neg\,\mathrm{Prov}[\mathrm{Th}T](\mathrm{gn}\,R_{\mathrm{Th}T}).$$

The sentence $\neg\,\mathrm{Prov}[\mathrm{Th}T](\mathrm{gn}\,R_{\mathrm{Th}T})$, i.e.,

$$\neg\exists x\,(\mathrm{Prf}[\mathrm{Th}T](x, \mathrm{gn}\,R_{\mathrm{Th}T}) \wedge \forall z\,(z < x \rightarrow \neg\,\mathrm{Refut}[\mathrm{Th}T](z, \mathrm{gn}\,R_{\mathrm{Th}T}))),$$

is logically equivalent to

$$\forall x\,(\mathrm{Prf}[\mathrm{Th}T](x, \mathrm{gn}\,R_{\mathrm{Th}T}) \rightarrow \exists z\,(z < x \wedge \mathrm{Refut}[\mathrm{Th}T](z, \mathrm{gn}\,R_{\mathrm{Th}T}))).$$

We argue informally using logic, making use of facts about what ThQ derives. Suppose x is arbitrary and $\mathrm{Prf}[\mathrm{Th}T](x, \mathrm{gn}\,R_{\mathrm{Th}T})$. We already know that $\mathrm{Th}T \vdash /R_{\mathrm{Th}T}$, and so for every k, $\mathrm{Th}Q \vdash \neg\,\mathrm{Prf}[\mathrm{Th}T](\overline{k}, \mathrm{gn}\,R_{\mathrm{Th}T})$. Thus, for every k it follows that $x \neq \overline{k}$. In particular, we have (a) that $x \neq \overline{n}$. We also have $\neg(x = \overline{0} \vee x = \overline{1} \vee \cdots \vee x = \overline{n-1})$ and so by ?? , (b) $\neg(x < \overline{n})$. By ?? , $\overline{n} < x$. Since $\mathrm{Th}Q \vdash \mathrm{Refut}[\mathrm{Th}T](\overline{n}, \mathrm{gn}\,R_{\mathrm{Th}T})$, we have $\overline{n} < x \wedge \mathrm{Refut}[\mathrm{Th}T](\overline{n}, \mathrm{gn}\,R_{\mathrm{Th}T})$, and from that $\exists z\,(z < x \wedge \mathrm{Refut}[\mathrm{Th}T](z, \mathrm{gn}\,R_{\mathrm{Th}T}))$. Since x was arbitrary we get, as required, that

$$\forall x\,(\mathrm{Prf}[\mathrm{Th}T](x, \mathrm{gn}\,R_{\mathrm{Th}T}) \rightarrow \exists z\,(z < x \wedge \mathrm{Refut}[\mathrm{Th}T](z, \mathrm{gn}\,R_{\mathrm{Th}T}))).$$

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